

Literature Answer: Markov Convexity Problems from 0706.0545

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Status. `literature_already_answered`. Two extracted open problems from Lee–Naor–Peres, *Trees and Markov convexity*, arXiv:0706.0545, are answered in the later paper Mendel–Naor, *Markov convexity and local rigidity of distorted metrics*, arXiv:0803.1697.

Source questions. In arXiv:0706.0545, Section “Some open problems” (PDF pp. 5–6), Problem 1 asks for the relation between Markov p -convexity and uniform p -convexity for general Banach spaces. The source had proved that p -uniform convexity implies Markov p -convexity and had a Banach-lattice converse up to exponents $q > p$, but left the general Banach-space relation unclear.

Problem 2 asks whether the metric-tree obstruction theorem extends to arbitrary metric spaces: if a metric space is not Markov p -convex for any finite p , must it contain arbitrarily large complete binary trees with distortion arbitrarily close to 1?

Supporting answers. Mendel–Naor arXiv:0803.1697 explicitly cites Lee–Naor–Peres as `LNP-markov-convex`. On PDF p. 3 it says that Lee–Naor–Peres asked whether the Banach converse is true and answers it positively. Theorem `convexity-coincide` in that paper states that a Banach space is p -convex if and only if it is Markov p -convex. The later Theorem `renorming` gives the quantitative equivalent renorming from a finite Markov p -convexity constant.

On PDF p. 4, the supporting paper also answers the general-metric analogue negatively. Its Theorem `doubling` constructs a metric space X which is not Markov p -convex for any finite p , yet

$$\lim_{n \rightarrow \infty} c_X(B_n) = \infty,$$

indeed with exponential lower growth for a doubling example. Therefore nontrivial Markov convexity is not forced by the failure of binary trees to embed with distortion arbitrarily close to 1, answering Problem 2 negatively.

Why this is not new. The supporting authors explicitly knew they were answering the Lee–Naor–Peres questions: the source paper is cited as `LNP-markov-convex`, the Banach converse is stated as a question from that paper and answered positively, and the arbitrary-metric analogue is stated as a question from that paper and answered negatively.

Scope. This packet records already-known literature answers to Problems 1 and 2 only. It does not address the broad unique-obstruction questions for planar graphs, hyperbolic-plane subsets, or products of trees in Problem 3, nor the lamplighter asymptotic and L_1 -embedding questions in Problem 4.

References.

- J. R. Lee, A. Naor, and Y. Peres, *Trees and Markov convexity*, arXiv:0706.0545.
- M. Mendel and A. Naor, *Markov convexity and local rigidity of distorted metrics*, arXiv:0803.1697.